One Configuration, Tuned per Dataset

November 4, 2026

The configuration. One model for all 8 datasets: inside and outside edges get two separate colors (their own weights in the GNN), trained with GLASS's recipe on GLASS's splits. On the synthetic datasets every shared node is replaced by one copy per subgraph. The real datasets run without copies, because their copies graphs do not fit in GPU memory.

What we ran, per dataset:

1. An Optuna search over aggregation, pooling, hidden size, layers, lr, dropouts, batch size, weight decay, norm, jumping knowledge, head, and (synthetic only) edge counts. Each trial is 3 seeds, scored by mean validation. Trials: hpo_neuro 80, hpo_metab and em_user 160, the rest 300. Weak trials were stopped after seed 0.

2. The 3 best trials on 10 seeds. The one with the best 10-seed validation is reported. Test was never used to choose.

	ours		published	
dataset	one config	edge colors	GLASS	GPEN
ppi_bp	55.3 ±0.9	59.9 ±0.9	61.9 ±2.2	**64.4** ±0.9
hpo_metab	60.5 ±2.8	61.1 ±2.3	61.4 ±1.6	**63.9** ±0.9
hpo_neuro	64.6 ±0.7	**69.6** ±0.9	68.5 ±1.6	69.1 ±0.6
em_user	82.9 ±7.3	73.1 ±6.0	88.8 ±1.9	**91.2** ±1.3
density	**98.9** ±1.3	94.8 ±3.2	93.0 ±2.8	95.6 ±1.6
cut_ratio	**95.2** ±2.0	89.1 ±4.0	93.5 ±1.9	93.6 ±1.6
coreness	86.7 ±5.2	86.5 ±5.4	84.0 ±2.8	**87.6** ±1.9
component	**100.0** ±0.0	**100.0** ±0.0	**100.0** ±0.0	**100.0** ±0.0

Table 1: Test micro-F1 (%), mean ± std over 10 seeds. One config: this run. Edge colors: the best model per dataset from report 10 (a different setup on each dataset). GLASS and GPEN: published numbers from the GPEN paper. Bold: the highest score in the row.

dataset	aggregation	pooling	hidden	layers
ppi_bp	mean	mean	256	3
hpo_metab	mean	max	64	2
hpo_neuro	mean	gated attention	128	2
em_user	attention	gated attention	32	2
density	attention	transformer	32	1
cut_ratio	GCN	mean	16	1
coreness	GCN	mean	64	2
component	sum	sum / $\sqrt{size}$	32	1

Table 2: The chosen trial per dataset. Hidden: width of each GNN layer. Layers: number of GNN layers.

Findings

- **Best on 3 of 4 synthetic datasets.** density 98.9 (GPEN 95.6), cut_ratio 95.2 (GPEN 93.6), component 100. coreness is 0.9 below GPEN and 2.7 above GLASS.
- **The search fixed cut_ratio and density.** Against report 10: cut_ratio +6.1, density +4.1, with smaller spread between seeds.
- **Worse on the real datasets, except em_user.** Against report 10: ppi_bp -4.6, hpo_neuro -5.0, hpo_metab -0.6. em_user rose by 9.8 but is still 5.9 below GLASS.
- **On ppi_bp, validation chose the worst finalist.** It scored 71.8 on validation and 55.3 on test. The other two finalists scored 70.2 on validation and 59.4 and 59.3 on test.
- **On hpo_neuro, validation is below test.** All 3 finalists score about 58 on validation and 64 on test. This dataset had the smallest search (80 trials).
- **On em_user, validation hardly predicts test.** Over all trials, the correlation between validation and test is 0.22 (0.74 to 0.98 on the other datasets, except ppi_bp at 0.56). The 3 finalists have almost the same validation (80.6 to 81.0) but test 79.8 to 85.5.